EE236: Electronic Devices Lab
Lab 9: N-channel MOSFET I-V Characteristics

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1 Aim of the experiment

- Obtain output and transfer characteristics of an N-channel enhancement type MOSFET (also called NMOS).
- Measure of trans-conductance and output resistance from the obtained characteristics.
- Investigate the effect of body bias on the characteristics of the NMOS.

2 Theory

The MOSFET is a commonly used transistor in both digital and analog circuits. It has four terminals:

- Gate (G)
- Drain (D)
- Source (S)
- Body (B)

The working principle of the device is that the voltage applied between its gate and source terminal controls the current through the source and drain terminals. The body terminal of a PMOS is connected to the highest voltage in the circuit, i.e., VDD, while that of an NMOS is connected to ground.

2.1 Operating Conditions

For an NMOS to be "ON", the applied gate-source voltage must be greater than the threshold voltage, i.e., $V_{GS} > V_T$. Otherwise, it is said to be "OFF" (or in the cut-off region).

2.2 Conditions for Different Operating Regions of an NMOS

When the NMOS is turned on:

- Linear Region: $V_{DS} < V_{GS} - V_T$
- Saturation Region: $V_{DS} \geq V_{GS} - V_T$

2.3 NMOS Current Equation (in Different Regions)

$$
\begin{aligned}
I_D &\approx 0 \quad \text{(Cut-off region)} \\
I_D &= \mu_n C_{ox} \frac{W}{L} V_{DS} \left(V_{GS} - V_T - \frac{V_{DS}}{2} \right) \quad \text{(Linear region)} \\
I_D &\approx \frac{1}{2} \mu_n C_{ox} \frac{W}{L} (V_{GS} - V_T)^2 (1 + \lambda V_{DS}) \quad \text{(Saturation region)}
\end{aligned}
$$

Where:

- μ_n = mobility of charge carriers (electrons in NMOS)
- C_{ox} = per unit area capacitance between the gate and body
- W, L = width and length of the channel respectively
- λ = channel length modulation factor

2.4 Trans-conductance and Output Resistance

The trans-conductance and the output resistance are given by:

$$
\begin{aligned}
g_m &= \frac{\partial I_D}{\partial V_{GS}} \bigg|_{\text{const}\, V_{DS}} \\
r_o &= \frac{\partial V_{DS}}{\partial I_D} \bigg|_{\text{const}\, V_{GS}}
\end{aligned}
$$

3 Components Necessary

- CD4007 IC
- 1k Potentiometer ×2
- Breadboard and connecting wire
- DC power supply

4 Part I : Transfer Characteristics (Linear)

4.1 Circuit

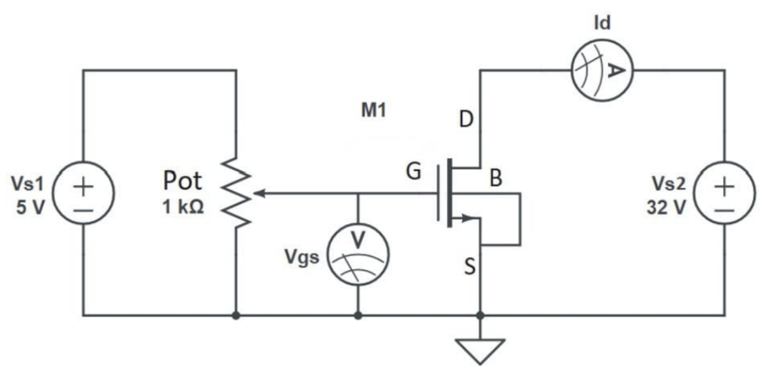

Figure 1: NMOS I-V circuit

4.2 Readings

V_{gs} (V)	I_d (A)
0	0.000
1.29	0.001
1.4	0.006
1.5	0.018
1.6	0.036
1.7	0.061
1.8	0.089
1.9	0.119
2	0.148
2.1	0.175
2.2	0.200
2.3	0.225
2.4	0.247
2.5	0.268
2.6	0.287
2.7	0.305
2.8	0.322
2.9	0.338
3.0	0.354

Table 1: Transfer characteristics in linear region

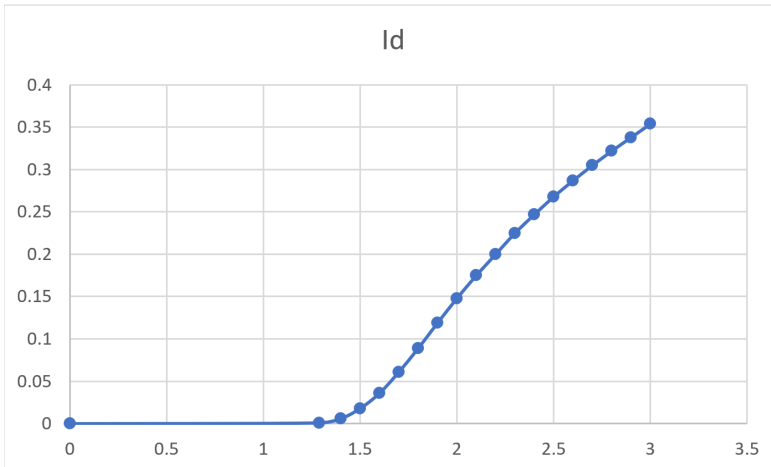

Figure 2: Linear Transfer Characteristics

- $V_T = 1.29$V
- $g_m = 0.227$ S

5 Transfer Characteristics (Saturation)

5.1 Circuit

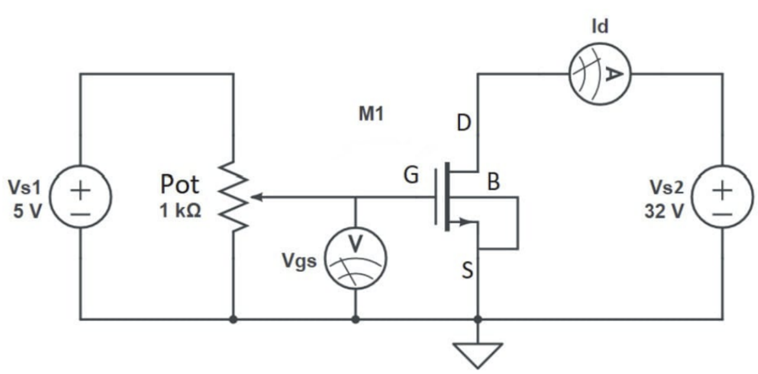

Figure 3: NMOS I-V circuit

5.2 Readings

V_{gs} **(V)**	I_d **(A)**
0	0.000
1.29	0.001
1.4	0.007
1.5	0.019
1.6	0.040
1.7	0.071
1.8	0.111
1.9	0.160
2.0	0.218
2.1	0.285
2.2	0.359
2.3	0.442
2.4	0.531
2.5	0.628
2.6	0.731
2.7	0.842
2.8	0.957
2.9	1.078
3.0	1.207

Table 2: Transfer characteristics in Saturation region

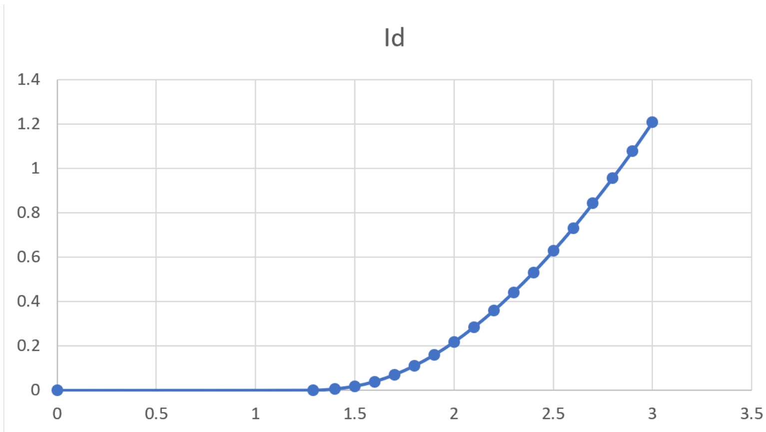

Figure 4: Saturation Transfer Characteristics

- $V_T = 1.29$V
- $g_m = 0.718$ S

6 Part II : Drain Characteristics

6.1 Circuit

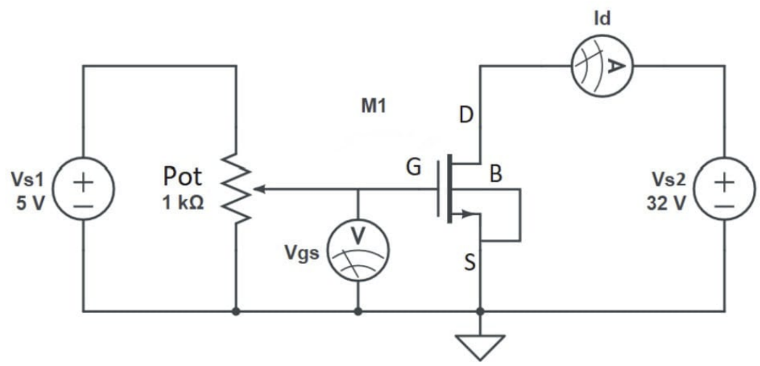

Figure 5: NMOS I-V circuit

6.2 Readings

V_{ds}	$I_d(V_{gs} = 1.5V)$	$I_d(V_{gs} = 2.5V)$	$I_d(V_{gs} = 3.5V)$
0.00	0.000	0.000	0.000
0.01	0.002	0.016	0.023
0.02	0.004	0.030	0.044
0.03	0.006	0.045	0.066
0.04	0.008	0.059	0.087
0.05	0.010	0.074	0.109
0.06	0.012	0.088	0.131
0.07	0.013	0.102	0.152
0.08	0.014	0.116	0.173
0.09	0.015	0.130	0.194
0.10	0.015	0.144	0.215
0.11	0.016	0.157	0.236
0.12	0.016	0.170	0.252
0.13	0.017	0.182	0.277
0.14	0.017	0.195	0.298
0.15	0.017	0.208	0.318
0.16	0.018	0.220	0.338
0.17	0.018	0.232	0.341
0.18	0.018	0.244	0.379
0.19	0.018	0.256	0.399
0.20	0.018	0.267	0.419
0.21	0.018	0.279	0.438
0.22	0.018	0.290	0.458
0.23	0.018	0.301	0.478
0.24	0.018	0.312	0.496
0.25	0.018	0.322	0.516
0.30	0.018	0.371	0.612
0.35	0.018	0.415	0.703
0.40	0.018	0.453	0.792
0.50	0.018	0.512	0.960
0.60	0.018	0.551	1.114
0.70	0.018	0.574	1.255
0.80	0.018	0.587	1.380
0.90	0.019	0.595	1.489
1.00	0.019	0.600	1.581
1.10	0.019	0.603	1.658
1.20	0.019	0.606	1.716
1.30	0.019	0.608	1.762

V_{ds}	$I_d(V_{gs} = 1.5V)$	$I_d(V_{gs} = 2.5V)$	$I_d(V_{gs} = 3.5V)$
1.40	0.019	0.610	1.795
1.50	0.019	0.612	1.818
1.60	0.019	0.613	1.837
1.70	0.019	0.615	1.851
1.80	0.019	0.616	1.862
1.90	0.019	0.617	1.870
2.00	0.019	0.618	1.875
2.10	0.019	0.619	1.882
2.20	0.019	0.620	1.886
2.30	0.019	0.621	1.892
2.40	0.019	0.622	1.895
2.50	0.019	0.623	1.898
2.60	0.019	0.624	1.901
2.70	0.019	0.624	1.905
2.80	0.019	0.625	1.908
2.90	0.019	0.626	1.910
3.00	0.019	0.627	1.911
3.10	0.019	0.628	1.914
3.20	0.019	0.628	1.916
3.30	0.019	0.629	1.918
3.40	0.019	0.630	1.921
3.50	0.019	0.630	1.924
3.60	0.019	0.631	1.926
3.70	0.019	0.631	1.927
3.80	0.019	0.632	1.929
3.90	0.019	0.633	1.932
4.00	0.019	0.633	1.933
4.10	0.019	0.634	1.933
4.20	0.019	0.634	1.935
4.30	0.019	0.635	1.938
4.40	0.019	0.636	1.939
4.50	0.019	0.636	1.941
4.60	0.019	0.637	1.942
4.70	0.019	0.637	1.944
4.80	0.019	0.637	1.945
4.90	0.019	0.638	1.946
5.00	0.020	0.638	1.946

Table 3: Caption

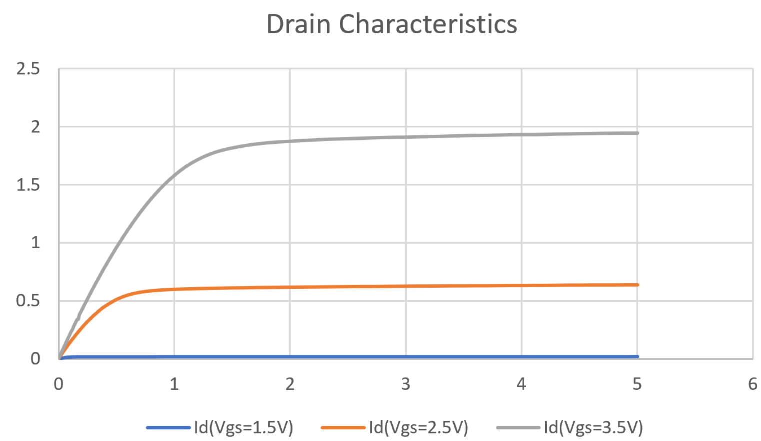

Figure 6: Drain Characteristics

- Slope of the plot for V_GS = 3.5V = 0.0288
- $r_o = 34.722k\Omega$

7 Part III : Body Effect

7.1 Circuit

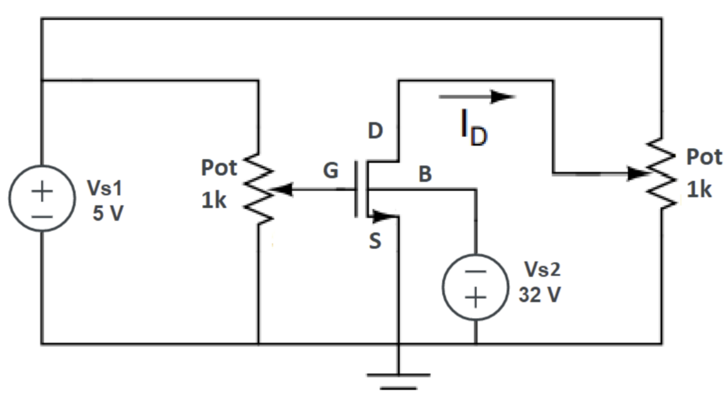

Figure 7: NMOS Body effect circuit

7.2 Readings

Vgs	Id(Vsb=0)	Id(Vsb=1V)	Id(Vsb=2V)	Id(Vsb=3V)
0	0	0	0	0
1.29	0.001	0	0	0
1.4	0.006	0	0	0
1.5	0.018	0	0	0
1.6	0.036	0	0	0
1.7	0.061	0	0	0
1.8	0.089	0	0	0
1.9	0.119	0	0	0
2	0.148	0	0	0
2.1	0.175	0	0	0
2.2	0.2	0	0	0
2.3	0.225	0	0	0
2.36	0.238	0.001	0	0
2.4	0.247	0.002	0	0
2.5	0.268	0.01	0	0
2.6	0.287	0.027	0	0
2.7	0.305	0.051	0	0
2.8	0.322	0.079	0	0
2.9	0.338	0.108	0	0
3	0.354	0.137	0	0
3.1		0.164	0	0
3.17		0.182	0.001	0
3.2		0.189	0.002	0
3.3		0.213	0.01	0
3.4		0.235	0.026	0
3.5		0.255	0.05	0
3.6		0.276	0.078	0
3.7		0.295	0.108	0
3.8		0.309	0.136	0
3.85		0.317	0.149	0.001
3.9		0.325	0.162	0.003
4		0.34	0.187	0.012

Vgs	Id(Vsb=0)	Id(Vsb=1V)	Id(Vsb=2V)	Id(Vsb=3V)
4.1			0.21	0.03
4.2			0.232	0.055
4.3			0.251	0.083
4.4			0.271	0.112
4.5			0.289	0.14
4.6			0.305	0.165
4.7			0.32	0.19
4.8			0.336	0.212
4.9			0.349	0.233
5			0.363	0.253
5.1				0.271
5.2				0.288
5.3				0.305
5.4				0.319
5.5				0.334
5.6				0.348
5.7				0.361
5.8				0.374
5.9				0.386
6				0.397

Table 4: Vgs and Id values for different Vsb

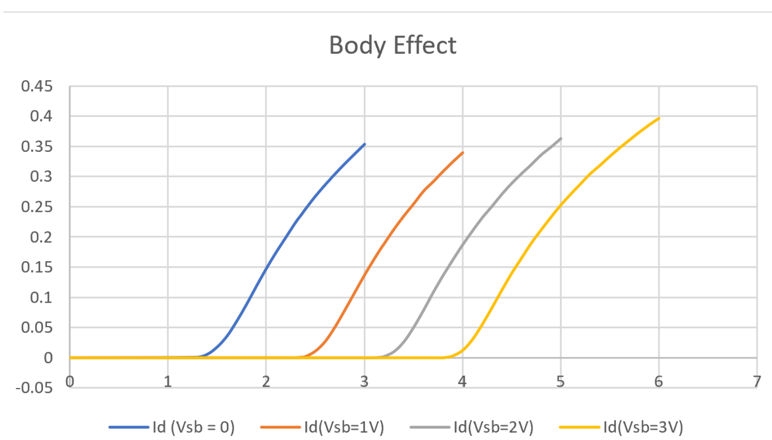

Figure 8: Body Effect

Using this expression,

$$V_T = V_{T0} + \gamma \left(\sqrt{\phi_s + V_{SB}} - \sqrt{\phi_s} \right) \tag{1}$$

V_{SB} (V)	V_T (V)	γ
0	1.29	0
1	2.36	2.4899
2	3.17	2.4925
3	3.85	2.4947

Table 5: Threshold voltage V_T and body effect parameter γ at different V_{SB} values.